

Coding & Solution

Date	30 JUNE 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/NarnamangalamBhumika/Credit-Card-Approval-Prediction-Code-File
Programming Language(s)	Python,HTML
Framework(s) Used	Flask,XGBoost,Scikit-learn
Key Features Implemented	Data preprocessing, Feature engineering, Label encoding, Feature scaling, Multiple ML model comparison (Logistic Regression, Decision Tree, Random Forest, XGBoost), Credit card approval prediction, Flask web application, Model deployment using Pickle

Pending / Incomplete Features	Cloud deployment, User authentication, Database integration, Real-time data integration
Setup / Run Instructions	1. Download or clone the repository. 2. Install dependencies using pip install -r requirements.txt. 3. Run the application using python app.py. 4. Open http://127.0.0.1:5000 in a web browser. 5. Enter applicant details and click Predict to view the prediction.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The project follows a modular approach with separate components for data preprocessing, model training, model serialization, and web application deployment. XGBoost was selected as the final model after comparing it with Logistic Regression, Decision Tree, and Random Forest due to its superior prediction performance. The application provides a user-friendly web interface for predicting credit card approval based on applicant details.